

Pratham Arora

+91-9810855594 | pratham3992@gmail.com | [linkedin.com/in/pratham3992arora](https://www.linkedin.com/in/pratham3992arora) | github.com/prats3992 | pratham-arora.vercel.app

Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India | July 2026

CGPA: 8.20 / 10.0

Industry Experience

Freelance Web Developer

May 2026 – Present

PQRS Research (Dr. Niket Tandon) | Remote

- Leading a full redesign of the PQRS Research website in Next.js, including a new dual onboarding system with distinct registration and intake flows for mentors and mentees.

AI Engineering Intern

June 2025 – July 2025

Cotality | Kolkata

- Architected a **FastAPI** backend with AST-based parsing for 15+ languages, enabling automated documentation generation across polyglot codebases at **50,000+ LOC/day**.
- Cut AI embedding costs by **85%** by building a hash-based system that fingerprints each function and skips re-processing unchanged code, eliminating redundant API calls on every pipeline run.
- Shipped **12 production REST API endpoints** with typed Pydantic schemas, powering real-time documentation preview in the client-facing Next.js interface.

Freelance Web Developer

May 2025

Anuj Desai Associates | Remote

- Delivered a full-stack site with a public careers page and a **Firebase**-backed admin dashboard, giving firm partners a private portal to review and manage job applications.
- Secured admin access via **Google OAuth** with an email allowlist (Firebase Auth), enforcing partner-only access without a separate auth backend.

Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud STT/TTS, achieving average response latency of **1.8s** through parallel API calls, caching, and audio preprocessing.
- Designed within-subjects user study ($N = 18$) evaluating three latency feedback mechanisms; findings ($p < 0.05$) directly informed the agent's final interaction design; **co-authored paper under review**.

AI Resume Builder & Interview Prep Tool | Next.js, Tailwind, Firebase, Gemini API

Jan. 2025 – Apr. 2025

- Built full-stack AI resume tool with real-time PDF preview and ATS-optimised generation using **Next.js** SSR and **Gemini 2.5 Flash** via streaming API.
- Integrated Gemini to generate personalised interview questions by cross-referencing uploaded resumes against a structured **DSA/STAR prep database**, surfacing role-specific preparation gaps.

Leadership & Involvement

Head of Technology

Apr. 2025 – Feb. 2026

Athleda – Plaksha Sports Society | Plaksha University

- Built and maintained a sports platform serving **500+ students**, shipping event registration, live-scoring, and photo galleries, while mentoring 2 junior developers through code reviews and deployments.

LEAP.AI @ Plaksha

Feb. 2024 – Feb. 2025

Web Developer → Head of Web Development | Plaksha University

- Led a team of 7–8 through **2 full platform redesigns**, growing the AI club site to **800+ monthly users** for events, workshops, and project showcases.

Skills

Languages: Python, SQL, JavaScript, TypeScript

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, Pydantic, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Firebase, Git/GitHub, Vercel